LENS · CONTENT REVIEWER & CONSISTENCY AUDITOR

Enterprise BFSI Study Guide & System Architecture Specification · Zero-Bug Blueprint

1. Executive Overview & Agent Identity

LENS is the 4th agent in the 10-agent Content Intelligence Platform (*CLARIA · LENS · INDEX · BRIDGE · GRID · ADAPTA · GLANCE · SORTIA · ECHO · ALLY*). While CLARIA redesigns customer communications and INDEX tags enterprise knowledge graphs, **LENS serves as the final gatekeeper and consistency auditor** before any content goes live.

Primary Question Answered: *"Is this content clear, complete, consistent and ready to publish?"*

Attribute	Specification
Agent Identity	LENS · Enterprise Content Reviewer & Consistency Auditor
Core Focus	Contradiction detection, conflicting dates/numbers, ambiguity, publish readiness
AI Model	Amazon Bedrock Nova Pro (us.amazon.nova-pro-v1:0) with Claude 3 Haiku fallback
Color Palette	Emerald Mint (#08d6a2) · Forest Emerald (#0b3d32) · Dark Panel (#06231b)
Review Output	0–100 Readiness Score · Actionable Review Matrix (Issue → Why Matters → Fix)

2. System Architecture & Dual-Engine Fallback

LENS follows a resilient serverless architecture built on AWS Lambda, Amazon Bedrock, Amazon S3, and CloudFront. To guarantee 100% uptime even during network or credential interruptions, LENS implements a three-tier execution hierarchy:

- **Tier 1 (Instant Conversational Matcher):** Intercepts greetings, typos (e.g. 'ji', 'gi', 'hlo'), and queries like 'how are you' with zero-latency contextual replies, preventing unnecessary LLM token spend.
- **Tier 2 (Bedrock Nova Pro Converse):** Analyzes cross-section semantics, identifies subtle timeline discrepancies, calculates readiness scores, and generates structured JSON audits.
- **Tier 3 (Heuristic Deterministic Fallback):** If Bedrock is unreachable, a local regex-powered contradiction engine parses numerical and date timelines, identifying day-count disparities without failing.

3. The Actionable Review Matrix Framework

A critical flaw in standard AI tools is the silent rewrite: LLMs often rewrite text without telling the author why. LENS adopts the enterprise standard **Issue → Why It Matters → Suggested Fix** matrix:

Component	Enterprise Purpose	Example

Location	Pinpoints exact sections where inconsistencies or flaws occur.	Section 2 vs Section 6
Severity	CRITICAL (blocker), WARNING (caution), SUGGESTION (polish).	**CRITICAL**
Verbatim Excerpt	Quotes exact wording from the document to establish provenance.	"within 30 days" vs "within 45 days"
Why It Matters	Articulates operational risk, legal exposure, or customer confusion.	Recipients receive contradictory SLAs, risking disputes.
Suggested Fix	Provides copy-paste replacement text ready for immediate insertion.	"within 30 calendar days of invoice date"

4. Zero-Bug Implementation Standards (Lessons from Taxora, Claria & Index)

During the development of previous agents, specific failure modes were documented. LENS integrates all preventive safeguards into its architecture:

Previous Bug Encountered	Root Cause	LENS Built-in Safeguard
Raw HTML in Chat Bubble	Missing VITE_API_BASE in production caused requests to hit CloudFront S3 fallback index.html.	deploy_aws.py automatically injects API Gateway URL into .env.production and triggers automated rebuild prior to S3 upload.
Bedrock ValidationException on 'ji'	User sent typo greeting 'ji'; Converse API requires strict role alternation starting with 'user'.	Fuzzy typo regex intercepts greetings immediately; Converse messages enforce strict role alternation starting with 'user'.
Duplicate 'Open Document' Badges	Both user bubble and AI answer card rendered document chips.	Single badge in user bubble only; AI cards display clean review metrics and copyable fix matrices.
Unscrollable Chat Page	Nested CSS with overflow: hidden on .messages and non-standard font weights (850, 650).	Pure standard multiples of 100 (400, 600, 700, 800) with overflow-y: auto and auto-scroll to last AI card.
Empty Conversations Saved	Opening new chat immediately persisted an empty session into localStorage.	Strict filter: sessions are persisted only when messages.length > 0.
Viewer 'No preview available'	Document viewer failed when view_url was empty.	Lambda streams PDF inline and converts Office/Text formats to HTML, with raw text fallback.

5. Verification Test Suite & Validation Commands

To verify that LENS is 100% bug-free across local and production environments, execute the following commands:

```
# 1. Test conversational greetings and fallback engine:
.\taxora\backend\venv\Scripts\python.exe lens\test_backend.py

# 2. Test local frontend development server:
cd lens\frontend && npm run build

# 3. Run automated AWS deployment (Lambda, S3, CloudFront):
.\taxora\backend\venv\Scripts\python.exe lens\deploy_aws.py
```

6. Conclusion & Roadmap

LENS establishes an uncompromising quality standard for enterprise document publishing. By enforcing consistency, eliminating contradictions, and offering copy-ready fixes, LENS safeguards organizations against operational confusion and legal exposure. With 4 out of 10 agents completed (TAXORA, CLARIA, INDEX, LENS), the Content Intelligence Platform continues towards comprehensive enterprise automation.